

Assignment 4: WiFi Sensing through Channel State Information (CSI)

Wireless Network

Full Marks: 20

Background: The objective of this assignment is to gauge the robustness of WiFi sensing. We collected WiFi CSI (Channel State Information) when students joined online classes from their devices and made various head movements such as “nodding”, “shaking”, “looking forward” and so on. In this assignment, you will gauge the effectiveness of WiFi sensing when there are some obstacles in the path between Tx and Rx, such as wood, glass, and so on. You will employ few shots learning concept to enhance WiFi sensing.

Download the “CSI_wild.zip” folder from the link: <https://zenodo.org/records/11551205>. The “CSI_wild.zip” folder contains various CSI dataset files. File with names *fiber_1*, *fiber_2*, and *fiber_3* should be ignored. The same applies to files named *wood_1*, *wood_2*, *wood_3* and *glass_1*, *glass_2*, *glass_3*. Utilize the dataset files with the following naming conventions (total 15 files) only and ignore other files:

P_26_fiber_4.csv ... P_30_fiber_4.csv,

P_26_wood_4.csv ... P_30_wood_4.csv,

P_26_glass_4.csv ... P_30_glass_4.csv.

Where $P_{26} \dots P_{30}$ defines the participant’s identity and *fiber*, *wood*, and *glass* are obstacles of width $0.5 \times 4in$ placed between the Access point and Receiving device during data collection. The CSI samples are labeled with 7 head gesticulations: 1) looking forward, 2) looking down, 3) looking up, 4) looking left, 5) looking right, 6) nodding, and 7) shaking.

Question: An optimal WiFi sensing solution adapts to varied experimental settings. In this case, variations in system settings refer to differences in participants and obstacle types. One of the optimal solution approaches is few-shot learning, in which a trained model is re-trained with “few shots”, i.e., a few re-training samples of unseen settings, participants, or locations during testing.

- a. Preprocess the CSI: For each subcarrier, there are 2 values real and imaginary indexed at even and odd indices, respectively. Read the CSI dataset files, remove the first 128 columns (CSI data from the legacy header), and remove the pilot and null subcarrier. Hint: The helper code is available in the provided file. [2]
- b. Compute the amplitude for each subcarrier using the equation $amplitudes = \sqrt{(imaginary^2 + real^2)}$. [2]

- c. Denoise the CSI samples with Hampel filter, 1-D wavelet transform filter, and savgol filter. Hint: The helper code is available in the provided file. [3]
- d. Create folders with filenames, for example, *P_26_glass_4*, and then create separate files for each head gesticulation and save them within the respective folder. Hint: The helper code is available in the provided file. [1]
- e. Preprocessing the data for training with few settings out strategy. Combine the CSI files of two materials for training and the CSI files of the remaining materials for testing. For example, combine “wood” and “fiber” for training and “glass” for testing. [2]
- f. Prepare the input data for training. Hint: The helper code is available in the provided file. [1]
- g. Train the model. The complete model implementation and training process code has been provided in the provided file. [2]
- h. Evaluate the trained model. Take a few samples from the available data, retrain the model, and test it with the remaining samples. Report the total number of retraining samples, testing samples, retraining time, testing time, and testing accuracy. Repeat the above process with different numbers of retraining samples. Does retraining help in improving accuracy? [4]
- i. Plot a graph where the X-axis represents total retaining samples, and the Y-axis represents accuracy. Discuss the trade-off between number of samples needed for retraining vs accuracy. [2 + 1]

Link of the helper code: <https://drive.google.com/drive/folders/1YR1e7rF3xnF-cQ92rKzhJhu0sS>
usp=sharing

Submission Instructions: Make a zip file with your code and report.

- Prepare a report with all necessary output.
- Submit your code